LAB 7 REPORT

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SECTION : J

1. Introduction

The objective of this lab was to perform fuzz testing on a Python shopping cart application using the Hypothesis library. Fuzz testing automatically generates unexpected and invalid inputs to identify errors in a program. In this lab, three functions were tested: **cart_total()**, **promo_percent()**, and **delivery_zone()**. The initial tests revealed input-validation bugs, which were then fixed to make the application more robust.

2. Software and Tools Used

The following tools were used to complete the lab:
• Python 3.12.4
• Hypothesis 6.170.15
• pytest 9.2.0
• Windows PowerShell and Python virtual environment (venv)

3. Functions Tested

Function	**Description**
cart_total()	Calculates the total price of items in the shopping cart.
promo_percent()	Extracts the discount percentage from a promo code such as OFF20.
delivery_zone()	Identifies the delivery zone using the first digit of a pincode.

4. Bugs Identified Using Hypothesis

The initial fuzz tests failed for all three functions. Hypothesis generated unexpected inputs that caused exceptions in the original application. Examples included **None**, empty strings, **':'**, **'OFF'**, and **'0'**.

Function	**Example input**	**Error observed**
cart_total()	None, empty string, ':'	AttributeError, ValueError
promo_percent()	None, empty string, 'OFF'	AttributeError, IndexError, ValueError

delivery_zone()	None, empty string, '0'	TypeError, IndexError, KeyError

5. Fixes Implemented

Input validation was added to all three functions. The **cart_total()** function now checks the cart format and returns 0 for invalid input. The **promo_percent()** function validates the promo code and returns 0 when the code is invalid. The **delivery_zone()** function checks that the pincode contains six digits and returns 'Unknown' for invalid input. These checks prevent the functions from crashing on the tested unexpected inputs.

6. Testing and Results

The corrected application was tested using the cart input **pen:10x2,book:250x1**, promo code **OFF20**, and pincode **560100**. The application displayed a cart total of **270**, discount percentage of **20**, and delivery zone of **South**. The final command **python -m pytest test_shopping_cart.py -v** completed successfully with **3 passed in 0.98s**.

7. Conclusion

This lab demonstrated how Hypothesis can be used to discover input-validation bugs by generating unexpected inputs. After adding defensive input checks to the three functions, all three fuzz tests passed. The shopping cart application is now more robust against the invalid inputs covered by the tests.

8. Screenshots

The following screenshots provide evidence of the initial fuzz testing failures, corrected application output, and successful final test execution.

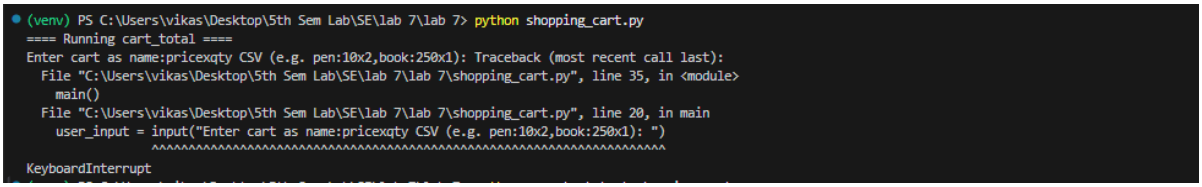


Figure 1: Hypothesis-generated failures in cart_total().

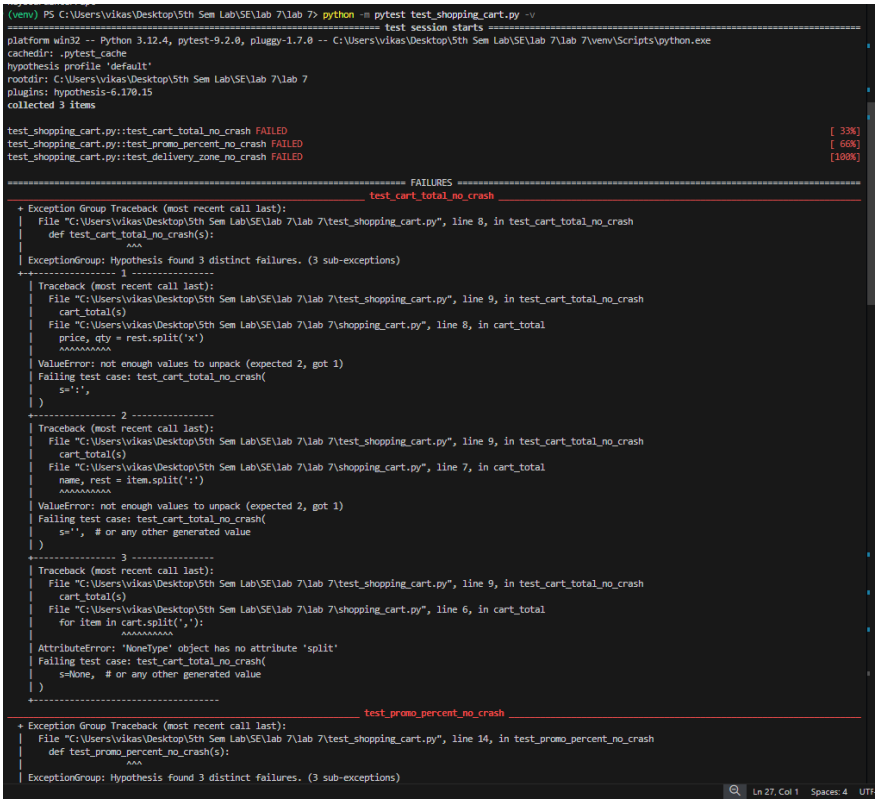


Figure 2: Hypothesis-generated failures in promo_percent().

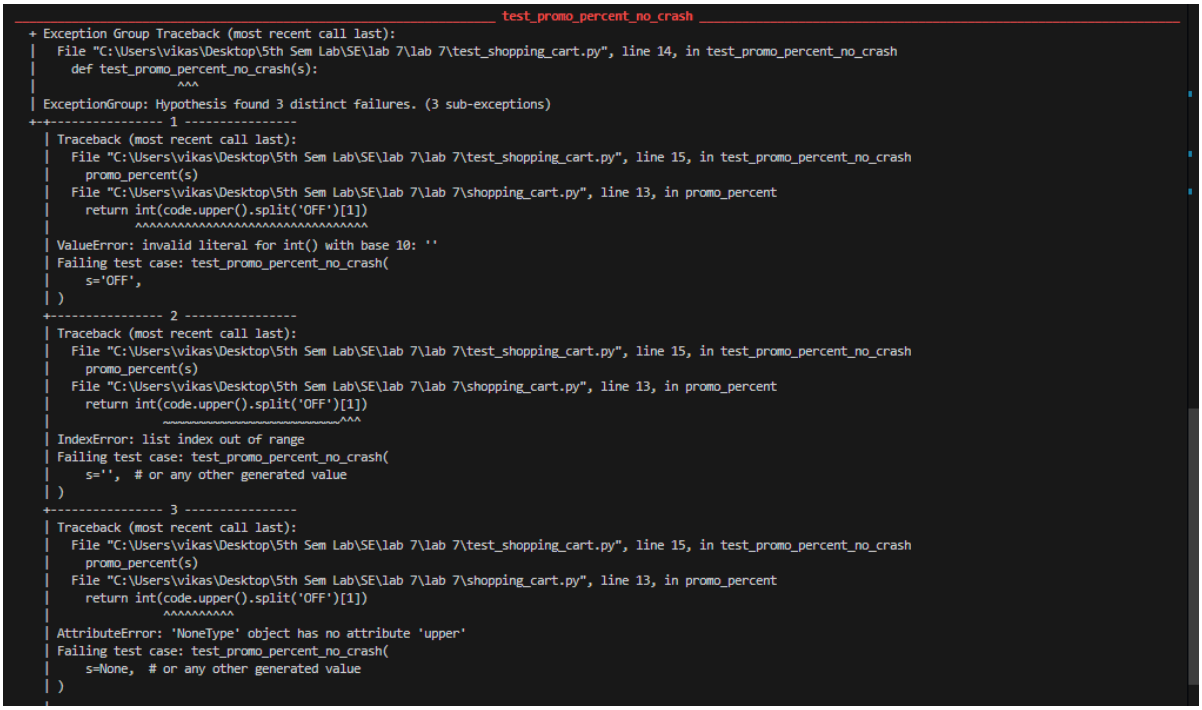


Figure 3: Hypothesis-generated failures in delivery_zone().

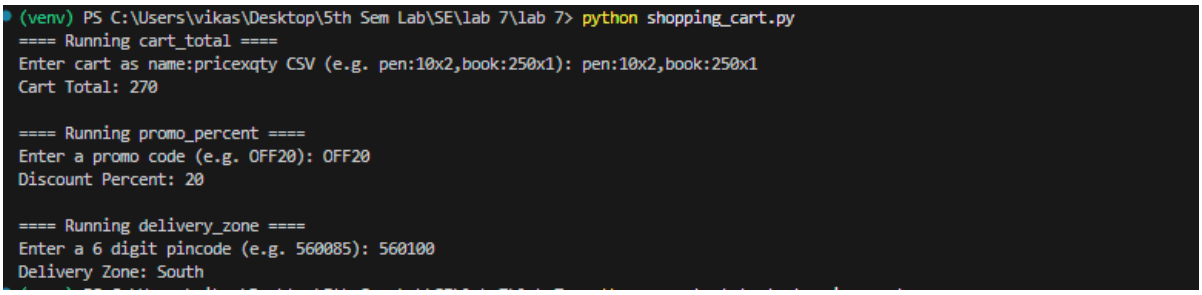


Figure 4: Output of the corrected shopping cart application.

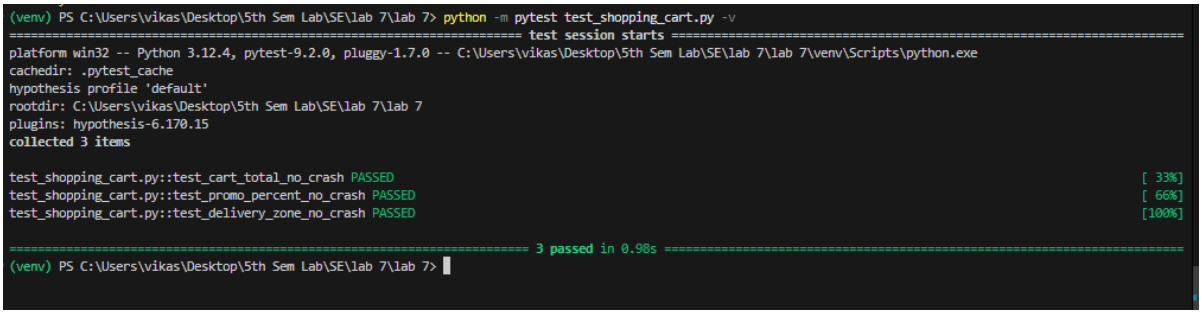


Figure 5: Final pytest execution showing all three tests passed.